

ISAIAH KAUNDA DZIMBIRI

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SUMMARY

Data Analyst with applied experience in machine learning, predictive modeling, and operational analytics across large and messy datasets. Built Random Forest, logistic regression, and k-means models on datasets exceeding 300,000 records, and delivered retail analytics (Pareto/ABC, market-basket, demand and inventory analysis) that surfaced measurable revenue and cross-sell opportunities for a multi-site business. Strong foundation in Python, R, SQL, and SQL database design, ETL, and dashboarding experience, plus six years of operations leadership. Owns the full workflow from data cleaning through modeling, visualization, and stakeholder-ready reporting.

PROFESSIONAL EXPERIENCE

Research Assistant, Data Analysis | Brandeis University

Feb 2026 – May 2026

- Cleaned, validated, and modeled large datasets in R and Excel; supported statistical analysis, sensitivity testing, and results interpretation.
- Produced analytical tables, figures, and written summaries translating technical output for non-technical academic and policy audiences.

Teaching Assistant | Brandeis University

Sep 2025 – Dec 2025

- Graded analytical assignments and prepared instructional materials on quantitative methods and data interpretation.

Managing Director and Pharmacist (Overseas) | Z+Plus Pharmacies

Mar 2019 – Aug 2025

- Led operations, data tracking, and performance reporting for a multi-site enterprise; supervised 7 to 12 staff and owned procurement, inventory, and financial reporting.
- Analyzed point-of-sale and inventory data to guide pricing, product mix, and purchasing decisions; built recurring Excel and dashboard reporting for management review.

Pharmacist and Manager (Overseas) | Mitch Pharmacies

Mar 2018 – Feb 2019

- Implemented workflow and reporting improvements that doubled sales within one year; managed inventory, procurement, and quality-assurance processes.

SELECTED DATA ANALYTICS PROJECTS

Retail Operations Analytics: Sales, Inventory, and Market-Basket Analysis

- Analyzed 6,110 point-of-sale records across 609 products in Python (pandas); applied Pareto/ABC classification, K-means clustering, and FP-Growth market-basket analysis.
- Found roughly 40% of period revenue came from out-of-stock products and 74% of invoices were single-item, quantifying concrete inventory and cross-sell opportunities.
- Delivered a multi-sheet Excel workbook and executive presentation with prioritized recommendations.

Predictive Modeling on a 300,000+ Record Dataset

- Co-developed Random Forest and logistic regression models to predict adverse outcomes; implemented an 80/20 train/test split and led hyperparameter optimization.
- Built parallel pipelines in Python (scikit-learn) and R; evaluated performance with ROC/AUC, confusion-matrix metrics, and variable-importance analysis.

Random Forest Classification for Performance Prediction

- Built and tuned a Random Forest model in R on 2,651 records (2,062 training / 589 test); selected mtry = 2 via 10-fold cross-validation.
- Achieved AUC 0.946, accuracy 87.1%, sensitivity 82.3%, specificity 90.3%; ranked predictors using Mean Decrease in Accuracy and Gini impurity.
- Segmented an additional 2,484-record dataset with k-means clustering (scikit-learn) and built a data-driven scoring and incentive scheme; measured market concentration with logistic regression, Origin-Destination matrices, and geographic mapping in R (sf).

EDUCATION

Brandeis University, Heller School for Social Policy and Management · MS, Global Health Policy and Management · May 2026

University of Malawi, College of Medicine · Bachelor of Pharmacy, Honors

TECHNICAL SKILLS

Programming & Statistics: Python (pandas, numpy, scikit-learn), R (caret, randomForest, ggplot2, sf), SQL, advanced Excel; OLS, panel data, IV, probit/logit, time series.

Machine Learning: Random Forest, logistic regression, k-means clustering, hyperparameter tuning, cross-validation, ROC/AUC, variable importance (Gini, MDA).

Business & Data Analytics: Pareto/ABC analysis, market-basket (FP-Growth) analysis, demand and inventory analysis, forecasting, KPI and performance reporting, dashboarding.

Databases, ETL & Visualization: MySQL, PostgreSQL, DuckDB; schema design, normalization, indexing, ETL; Tableau, ggplot2, Excel dashboards.

Certifications: CITI Human Subjects Research and Responsible Conduct of Research (2025); Global Health Leadership and Management, Washington University (2024).